

NGOSA GATHAMS
Msc Computer Science and IT
Msc Cyber Security

Presentation outline

- Introduction
- Definitions
- Characteristics of cloud computing
- Types of cloud computing
- Service models
- Cloud computing framework.

INTRODUCTION

Scenario A

Imagine you have enough money and you need a place to live for the next five years. You have two options:

1. Rent an apartment (monthly payments, landlord handles repairs, you can leave anytime with short notice) or
2. buy a house (large down payment, you pay for all maintenance, you build equity but selling is slow and expensive)

What will be the best option? Explain why.

Introduction

Scenario 2

You need to travel around a busy city for one year. You can either

1. use a ride-share service (pay per trip, no parking or maintenance, but surge pricing at peak hours) or
2. buy a car (high upfront cost, you pay insurance and fuel, but the car is always ready and costs the same regardless of demand).
   If your trips are completely unpredictable – some days zero trips, other days 20 trips – which option is better?
   If you commute the same 10 km every weekday at 8 AM and 6 PM, which option is better?

Introduction

- **Scenario 3**

You need to eat dinner every night for a month. You can either

1. eat at a restaurant (no grocery shopping, no cooking, no cleanup, but you pay per meal and the menu is fixed) or
2. cook at home (you buy all ingredients, spend time preparing and cleaning, but you control exactly what you eat and the cost per meal is much lower if you cook in bulk).

Imagine you are a single student who eats alone and likes variety. Which option will be best? explain

Now imagine you are a family of six who eats the same staple meals every day. Which option suits each?

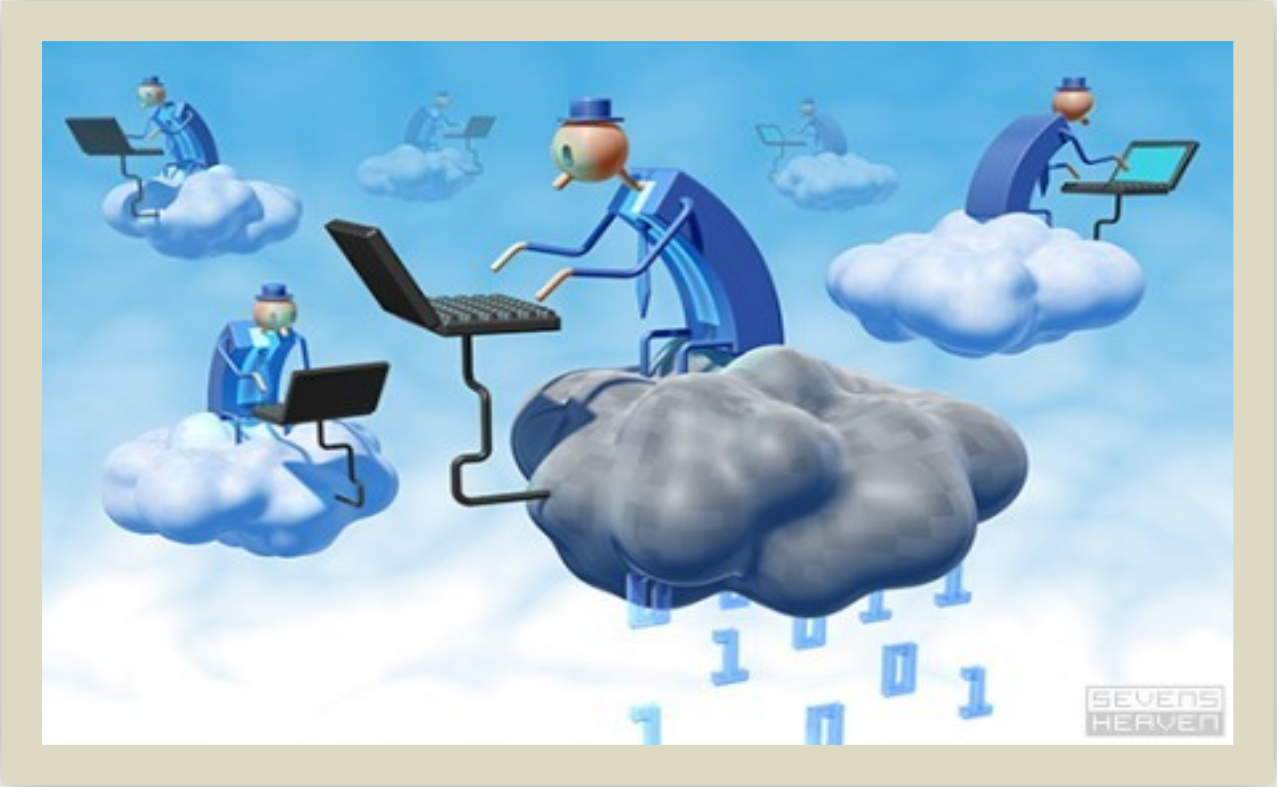
Introduction

- **Scenario 4**

You need to have only three kids in your life. There two options, to marry or have them with different partners.

Which option will be costly?

Cloud computing



DEFINITION

Cloud computing is a technology that uses the internet and central remote servers to maintain data and applications.

Cloud computing allows consumers and businesses to use applications without installation and access their personal files at any computer with internet access.

Five Essential Characteristics of cloud computing

1. **On-demand self-service**

A user can provision computing capabilities (e.g., server time, storage) automatically without requiring human interaction with the provider.

2. **Broad network access**

Resources are available over the network (typically the internet) and accessed through standard protocols (HTTP, HTTPS, SSH) by heterogeneous devices (phones, laptops, workstations).

Five Essential Characteristics

3 **Resource pooling**

The provider's computing resources are pooled to serve multiple tenants using a multi- tenant model, with physical and virtual resources dynamically assigned according to demand. The customer generally has no control over the exact location of resources (e.g., country, data centre).

4. **Rapid elasticity**

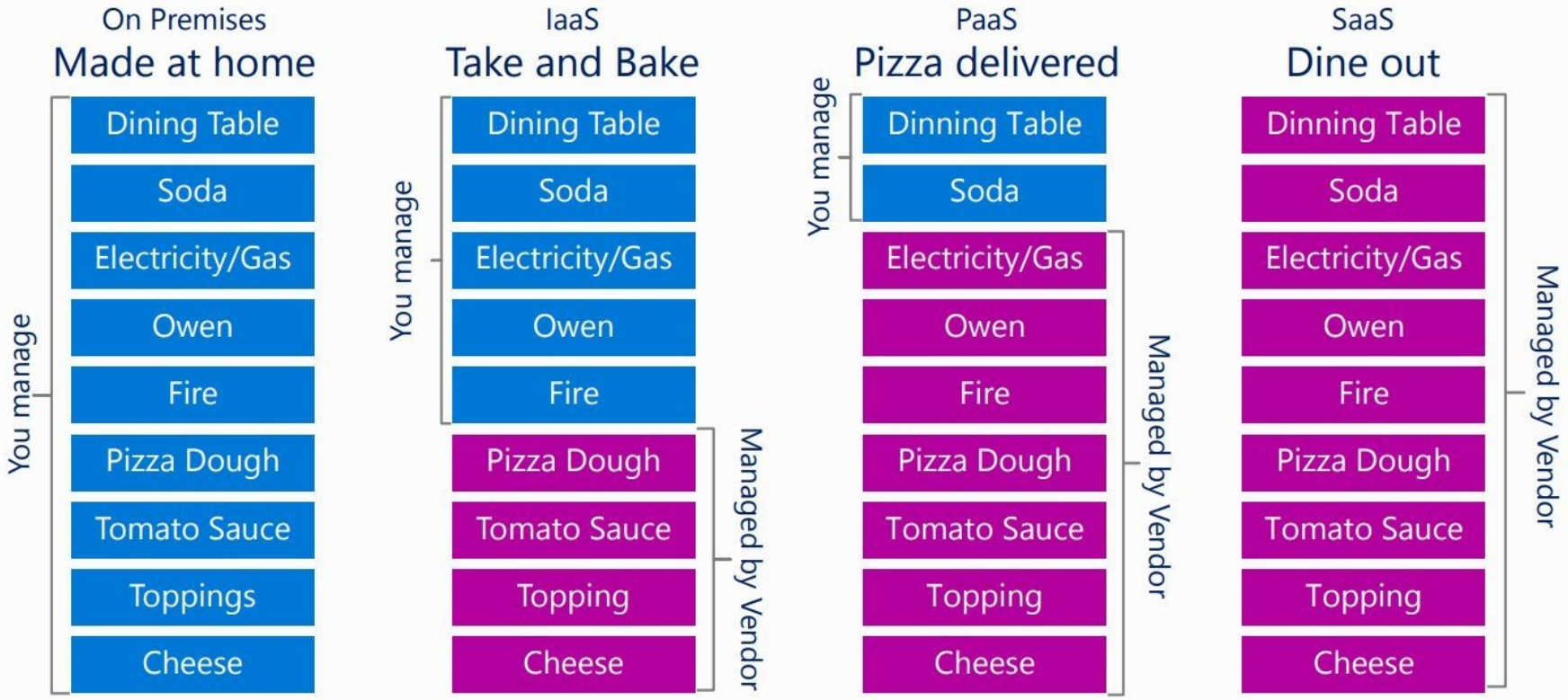
Capabilities can be elastically provisioned and released – in some cases automatically – to scale rapidly outward and inward commensurate with demand. To the user, the available resources often appear unlimited

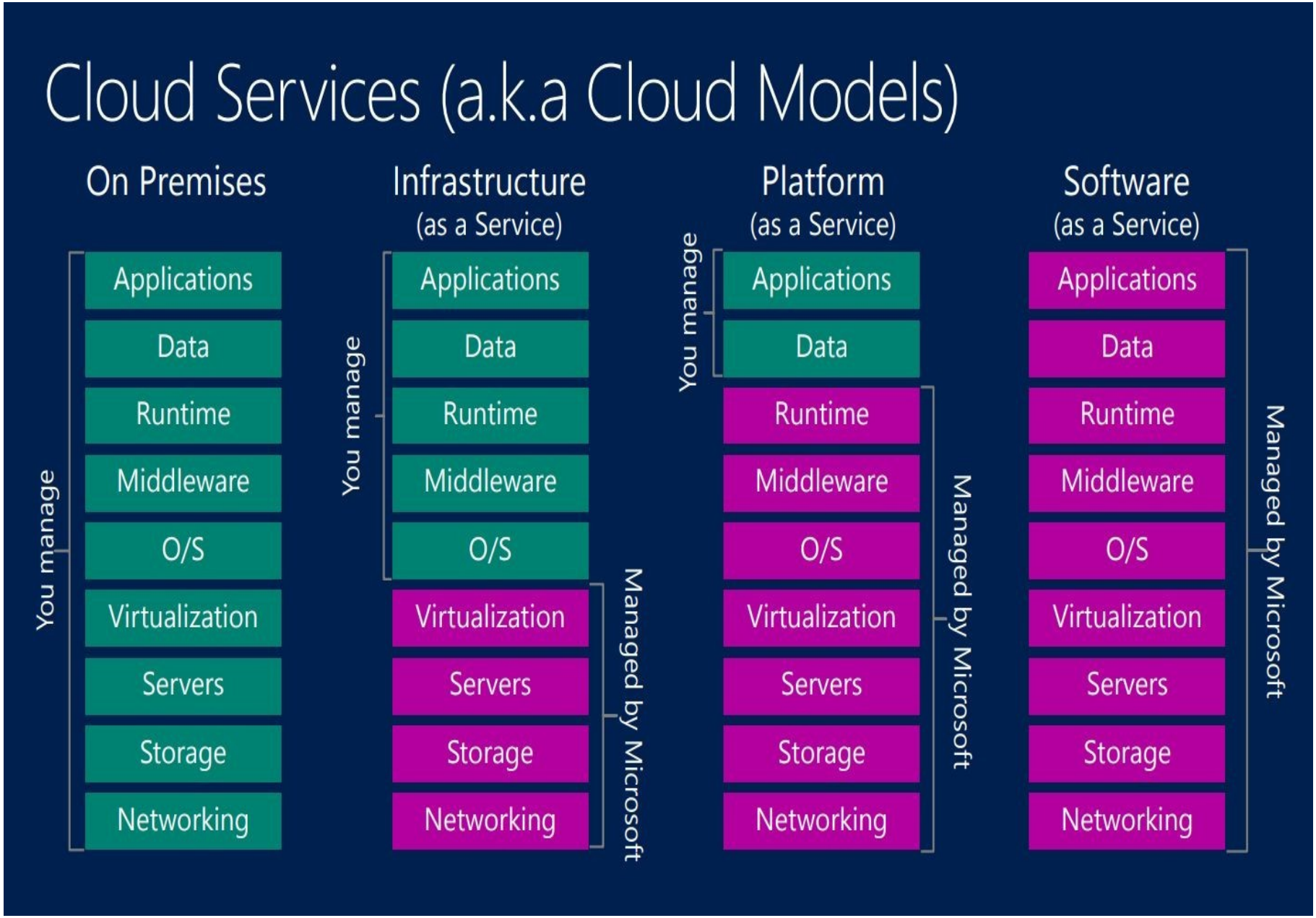
Five essential characteristics

5. **Measured service**

Cloud systems automatically control and optimise resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth). Usage is monitored, controlled, and reported – providing transparency for both provider and consumer.

Cloud Services *being hungry!*




Cloud Services (a.k.a Cloud Models)
On Premises
Infrastructure
(as a Service)
Platform
(as a Service)
Software
(as a Service)
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
You manage
Managed by Microsoft

Service Models

Cloud Computing Service Models are Geared for Different Purposes

Service Model	Who Uses It	Available Services	Why Use It
SaaS	Agency Staff Members	Software applications such as email, word processing, and customer relation management tools	Complete business tasks that are typically performed locally on a computer
PaaS	Developers and Application Managers	Services for creating, testing, managing, and hosting software applications	Establish a common and consistent platform for application development
IaaS	IT System Managers	Virtual machines, storage services, and backup services	Build a customized computing environment

The cloud computing reference model

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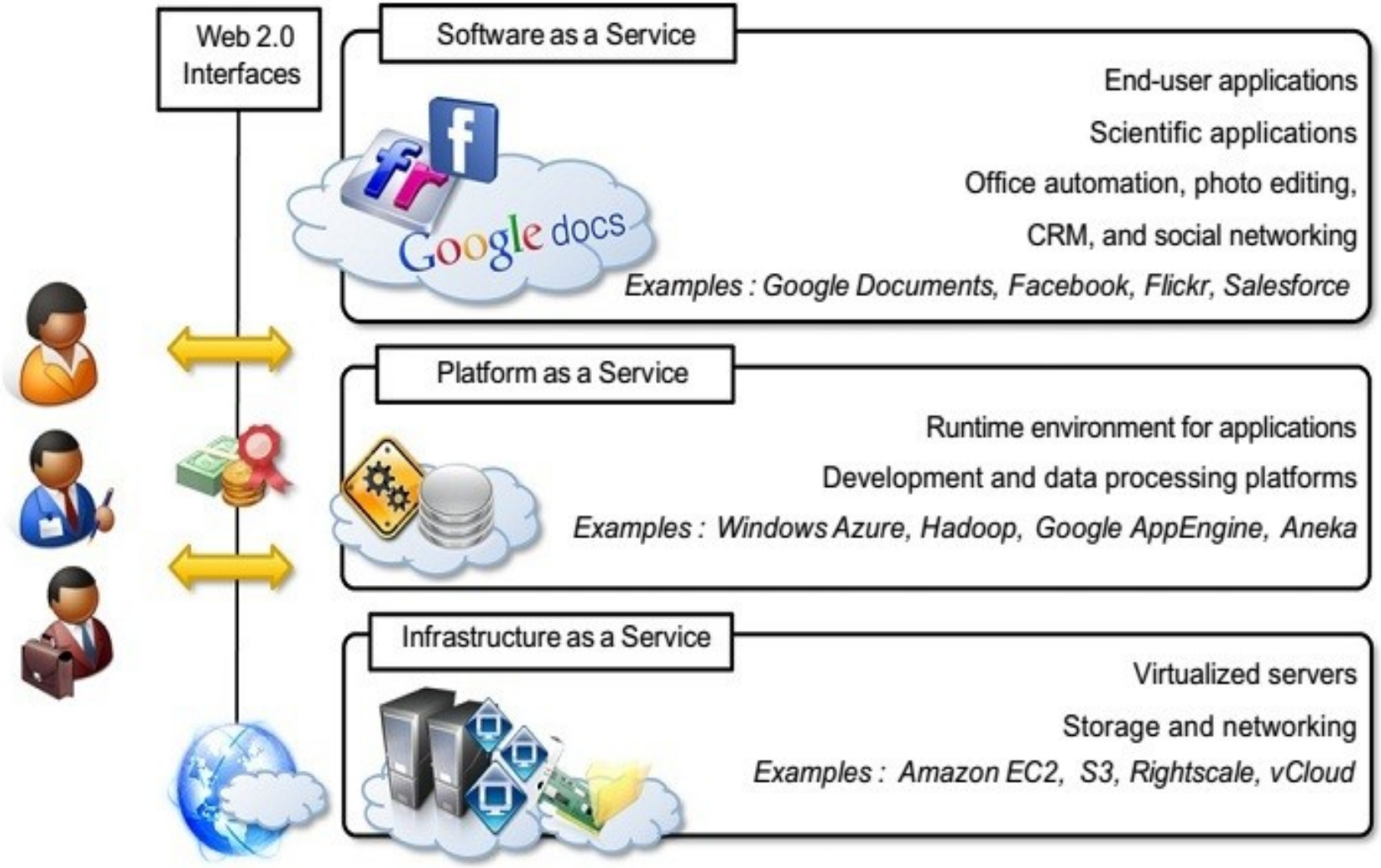


FIGURE 1.5

The Cloud Computing Reference Model.

TYPES OF CLOUDS

1. Public clouds
2. Private clouds.
3. Hybrid or heterogeneous clouds.
4. Community clouds.

1. **Public clouds.** The cloud is open to the wider public.
2. **Private clouds.** The cloud is implemented within the private premises of an institution and generally made accessible to the members of the institution or a subset of them.
3. **Hybrid or heterogeneous clouds.** The cloud is a combination of the two previous solutions and most likely identifies a private cloud that has been augmented with resources or services hosted in a public cloud.
4. **Community clouds.** The cloud is characterized by a multi-administrative domain involving different deployment models (public, private, and hybrid), and it is specifically designed to address the needs of a specific industry.

Hybrid clouds

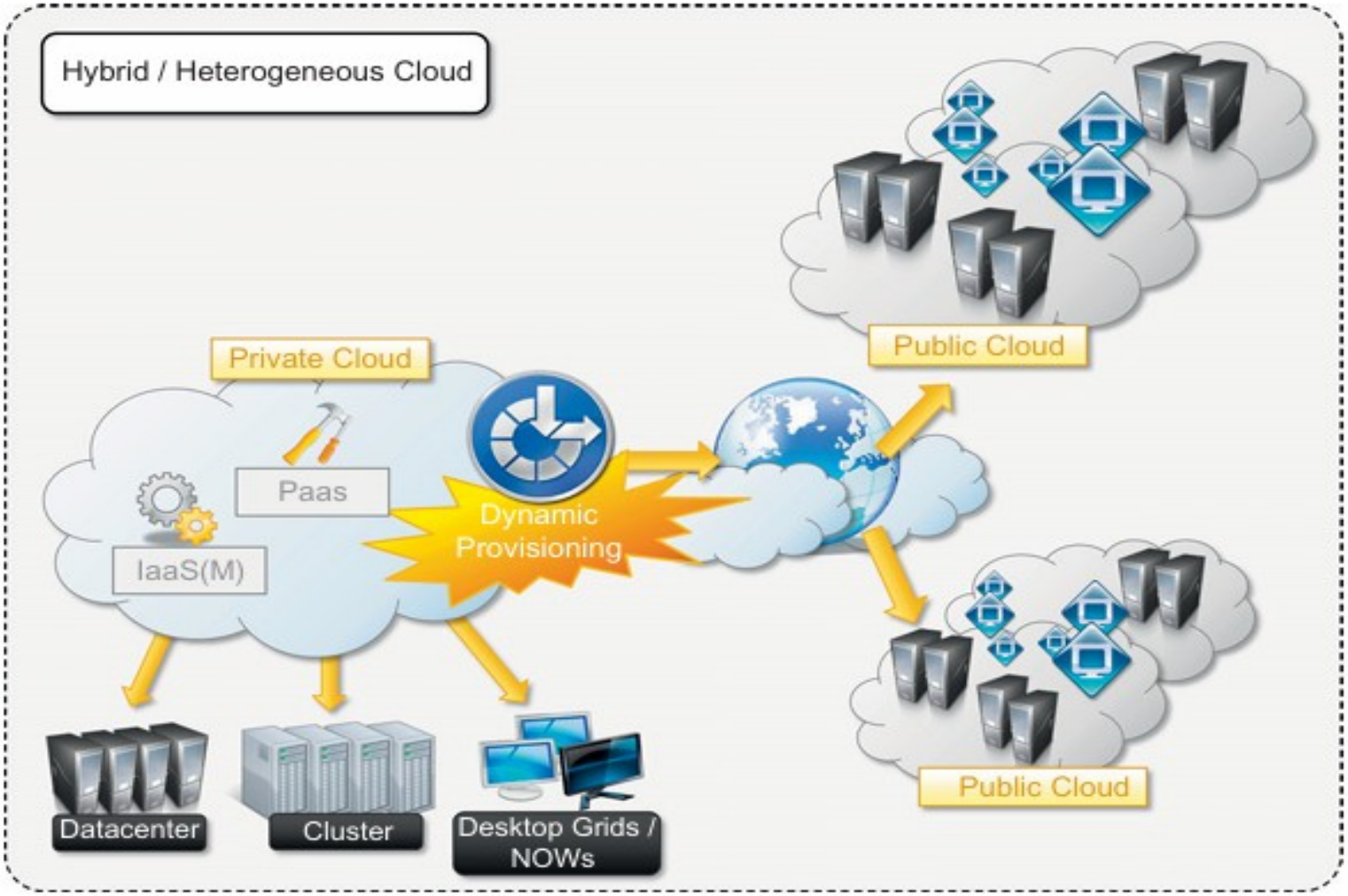


FIGURE 4.5

Hybrid/heterogeneous cloud overview.

Community clouds

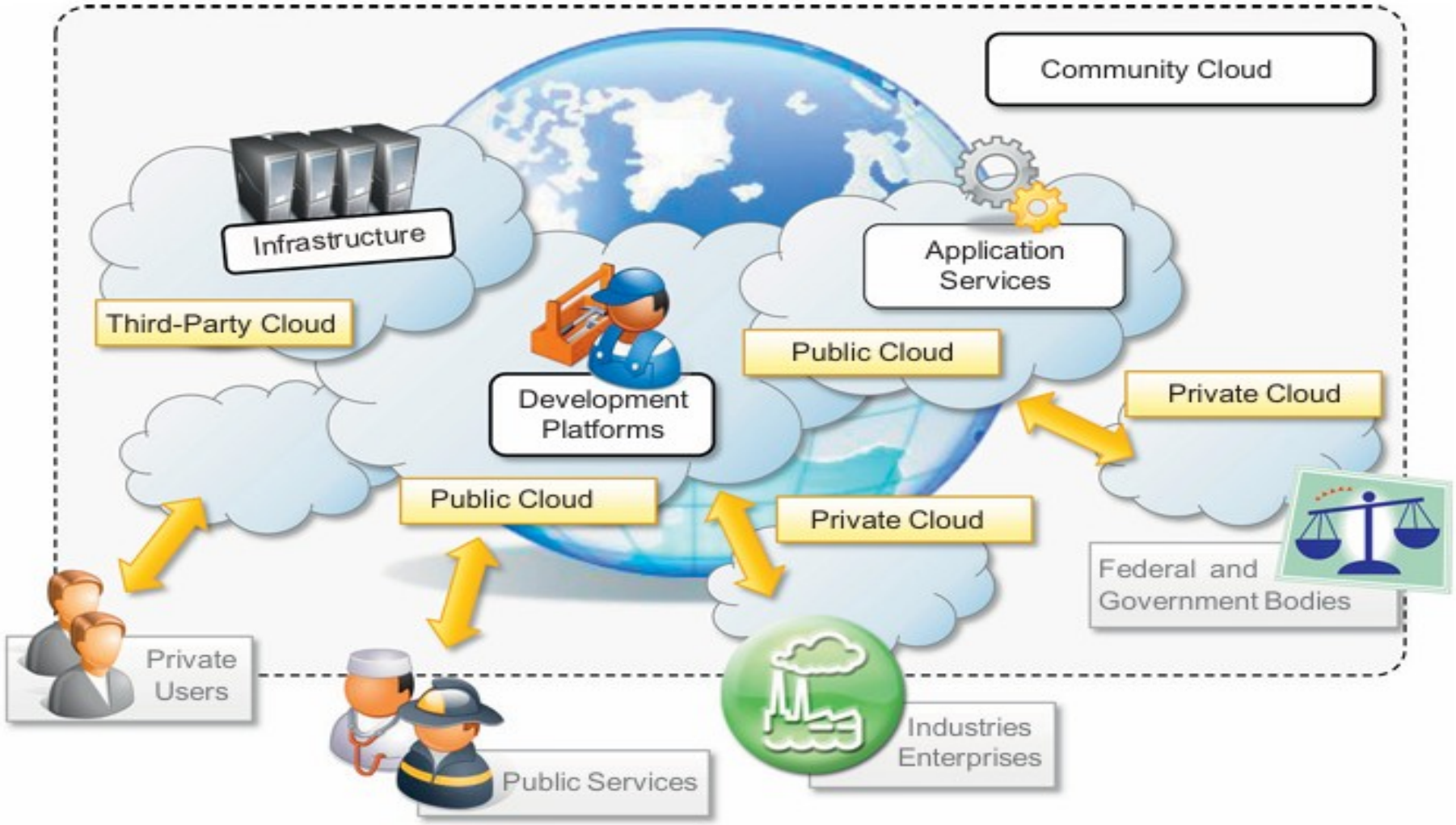
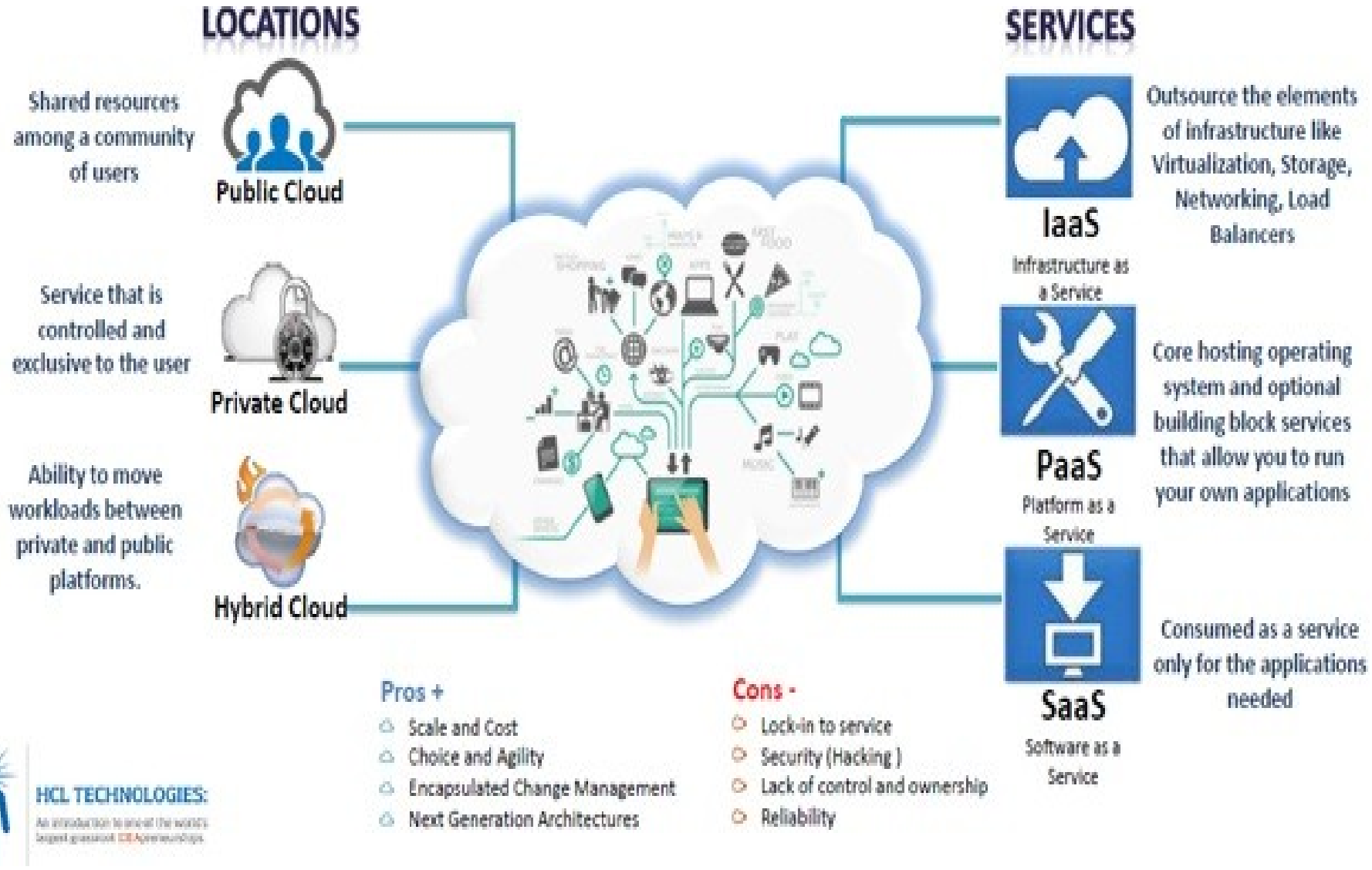


FIGURE 4.6

A community cloud.

Locations & Services

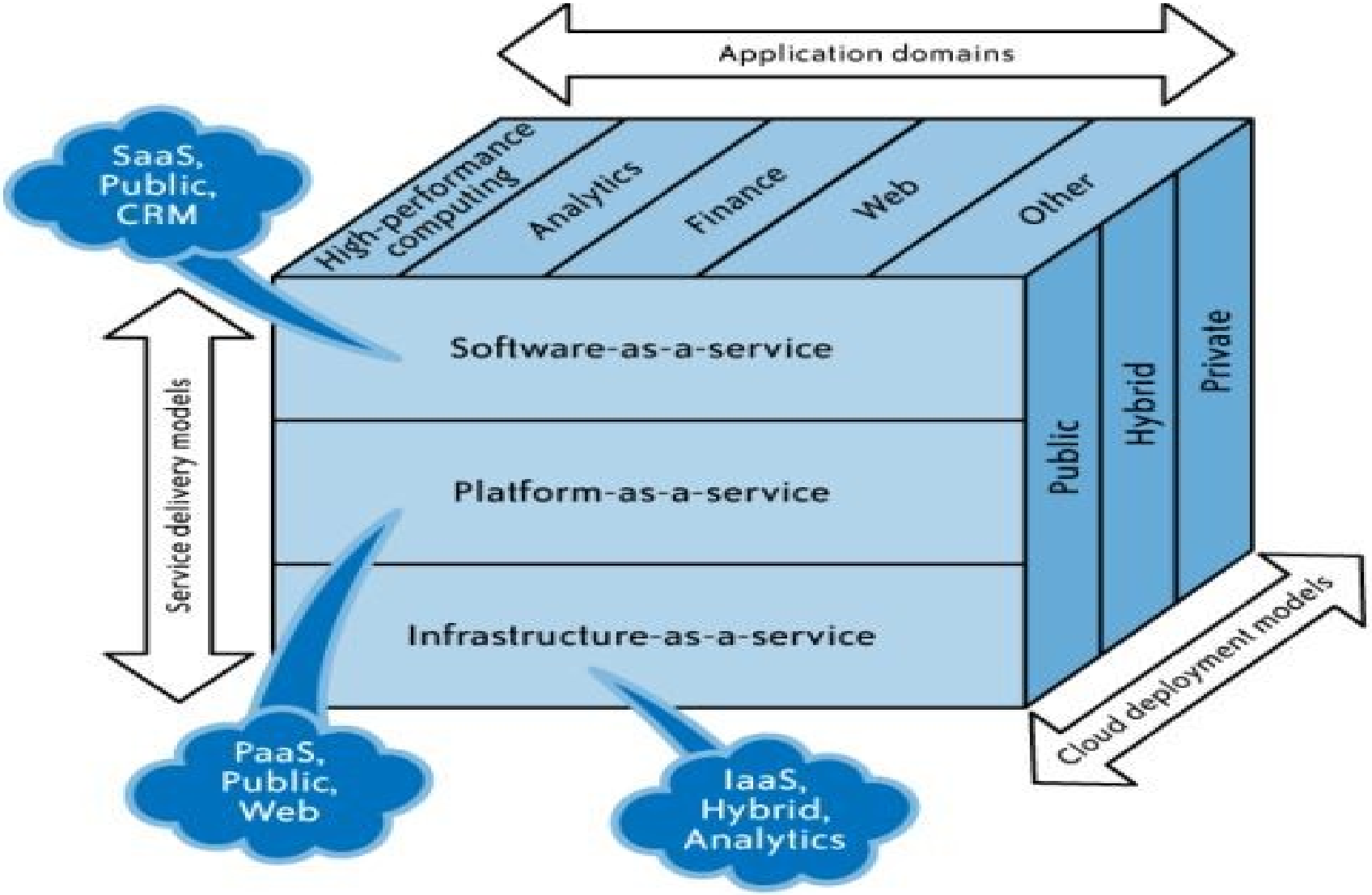


HCL TECHNOLOGIES:

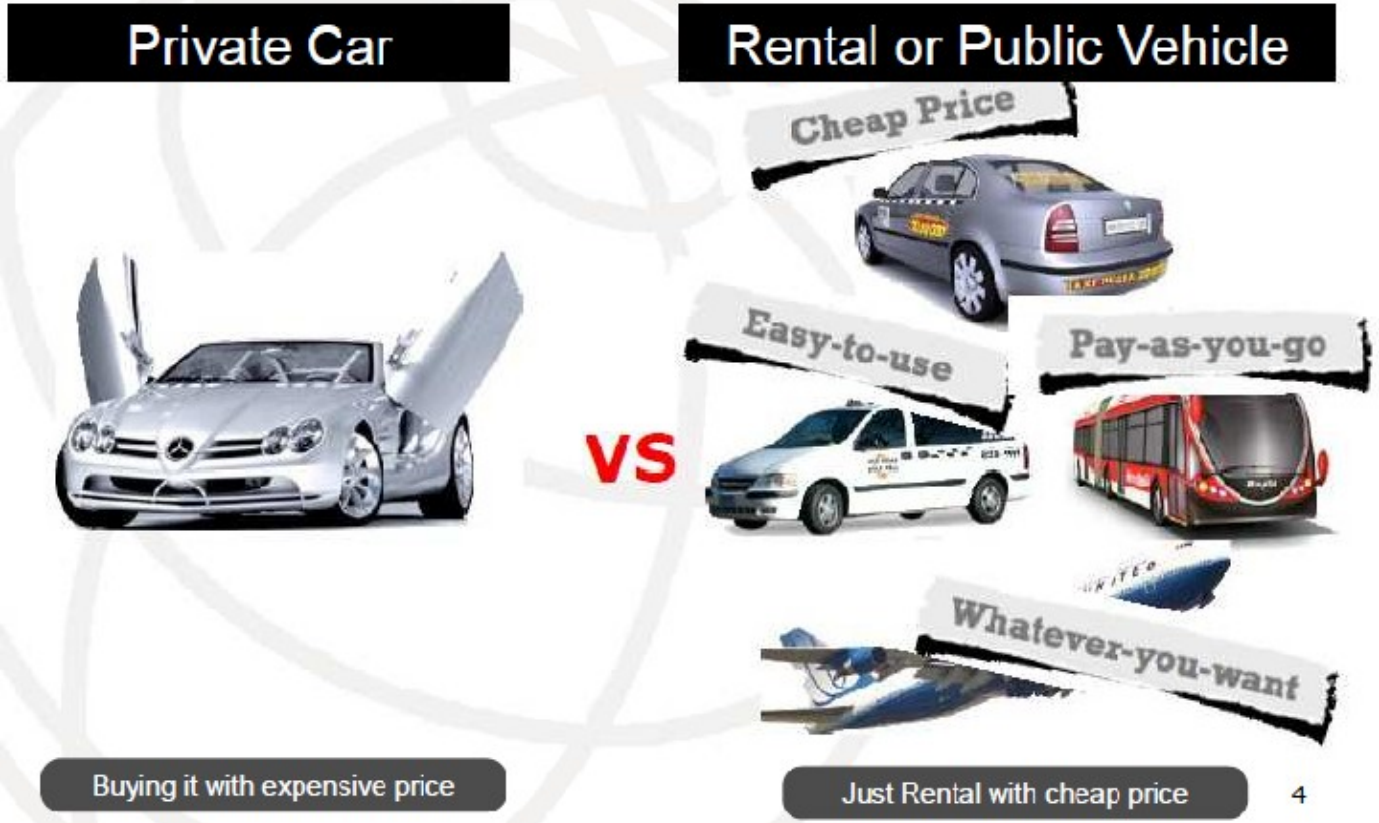
Deployment Models or cloud types

	Infrastructure Managed By[1]	Infrastructure Owned By[2]	Infrastructure Located[3]	Accessible and Consumed By[4]
Public	Third Party Provider	Third Party Provider	Off-Premise	Untrusted
Private/ Community	Organization Or Third Party Provider	Organization Third Party Provider	On-Premise Off-Premise	Trusted
Hybrid	Both Organization & Third Party Provider	Both Organization & Third Party Provider	Both On-Premise & Off-Premise	Trusted & Untrusted

Cloud Computing Framework

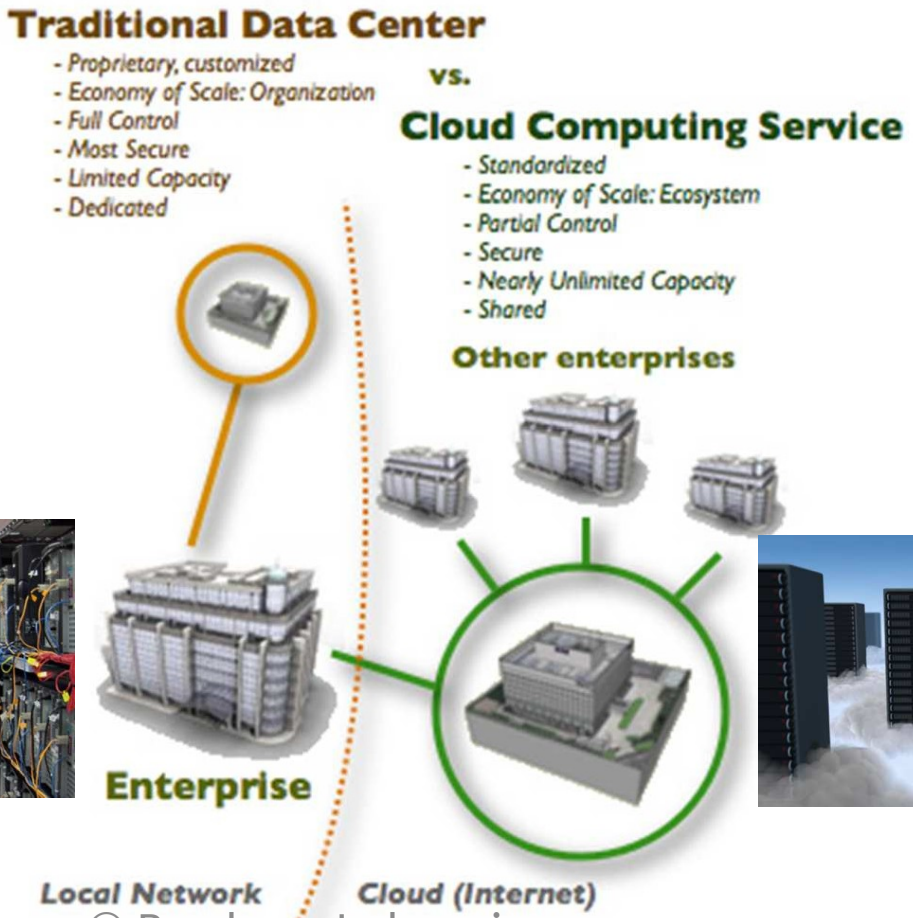


ESSENTIAL CHARACTERISTICS OF CLOUD



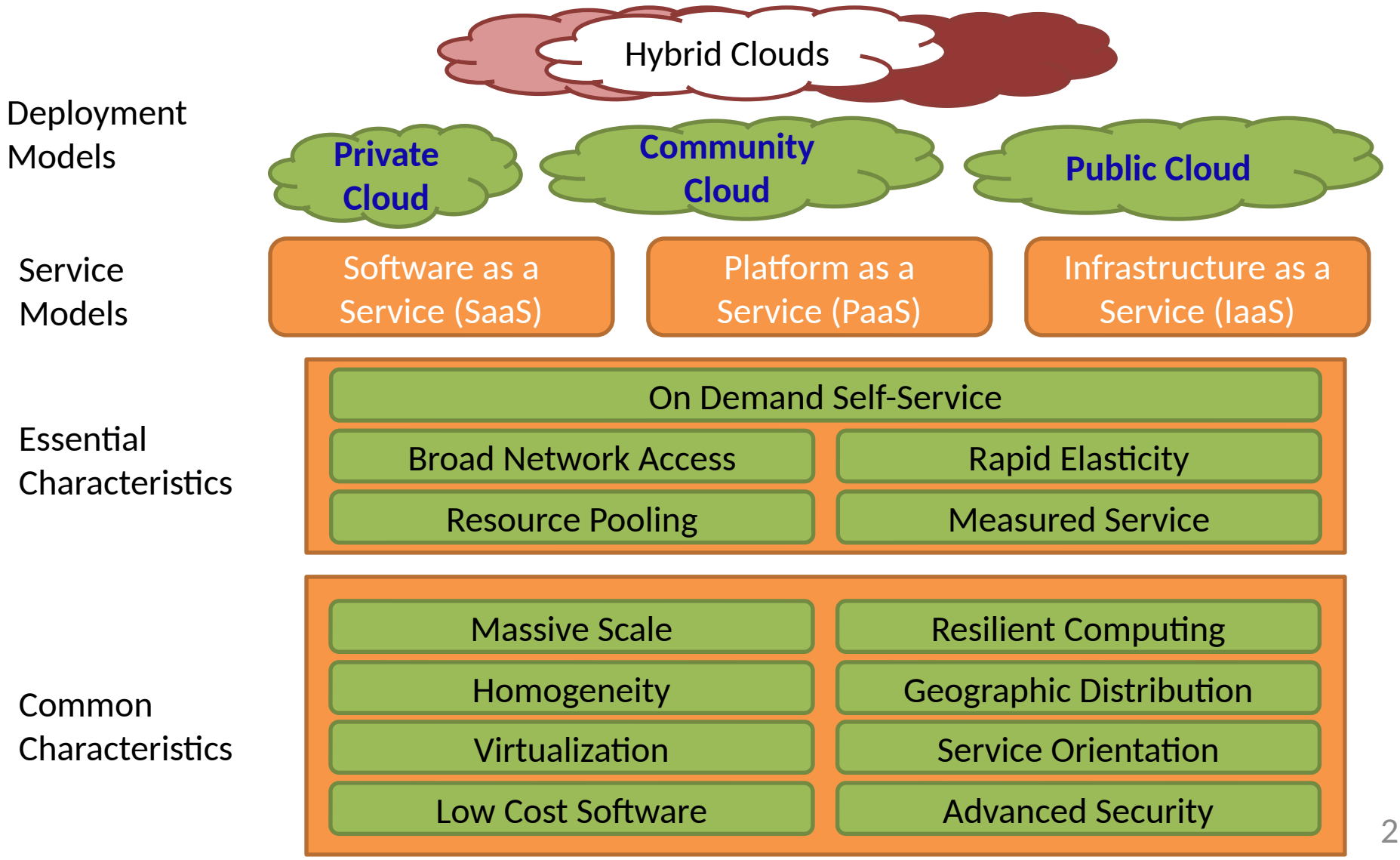
UNDERSTANDING THE DIFERENCES

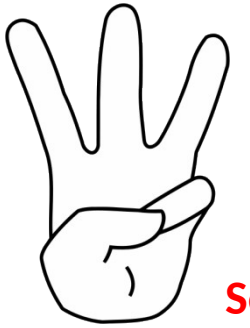






Cloud Computing Framework





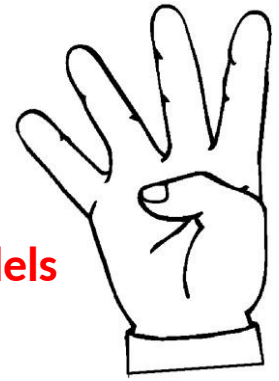
Service Models

- *SaaS*
- *PaaS*
- *IaaS*



- *Private*
- *Public*
- *Community*
- *Hybrid*

Deployment Models





Essential Services

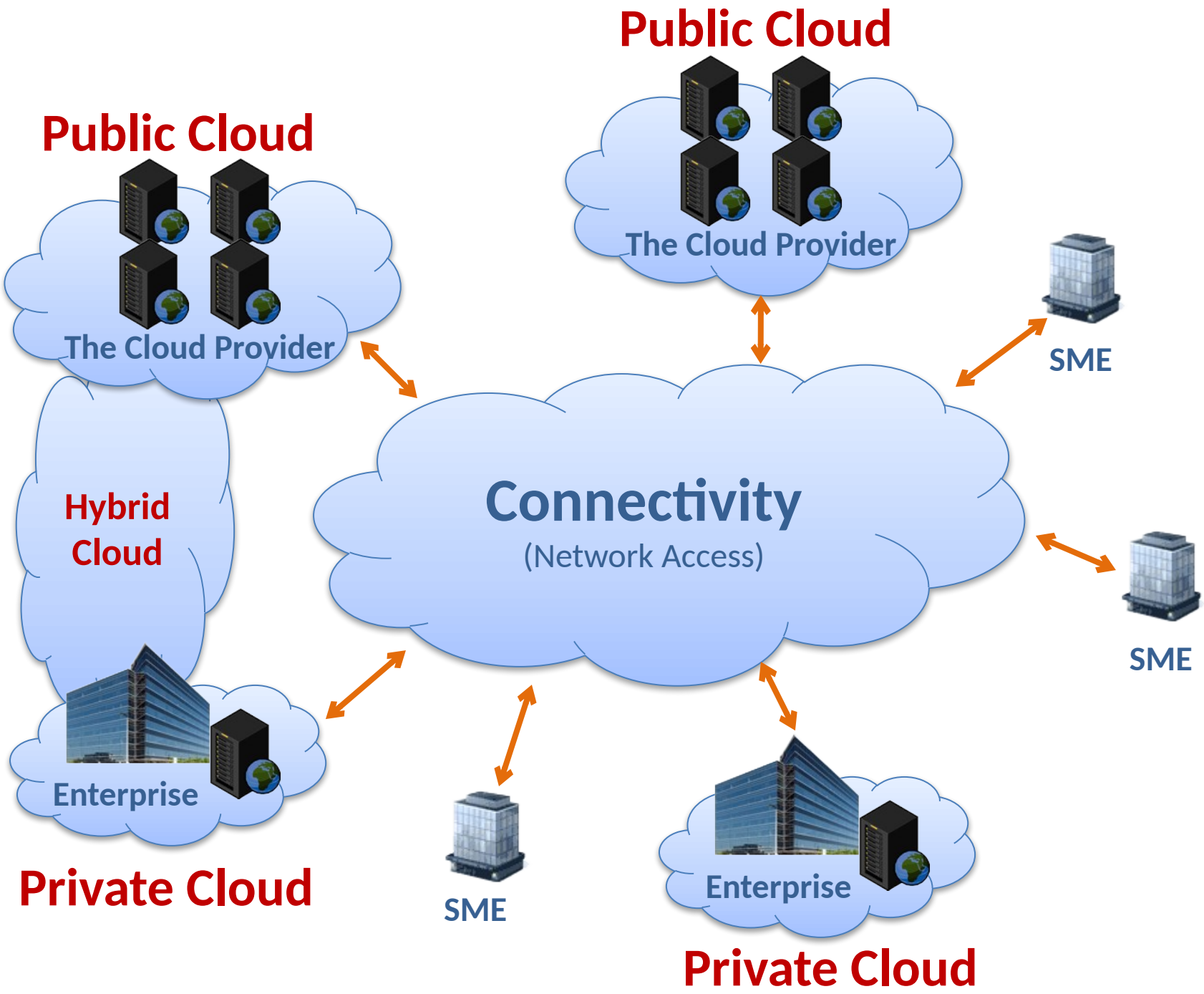
- *On-demand self service*
- *Broad network access*
- *Resource pooling*
- *Rapid elasticity*
- *Measured service*

Definitions

Cloud computing is a style of computing in which dynamically scalable and often virtualized resources are provided as a service over the Internet.

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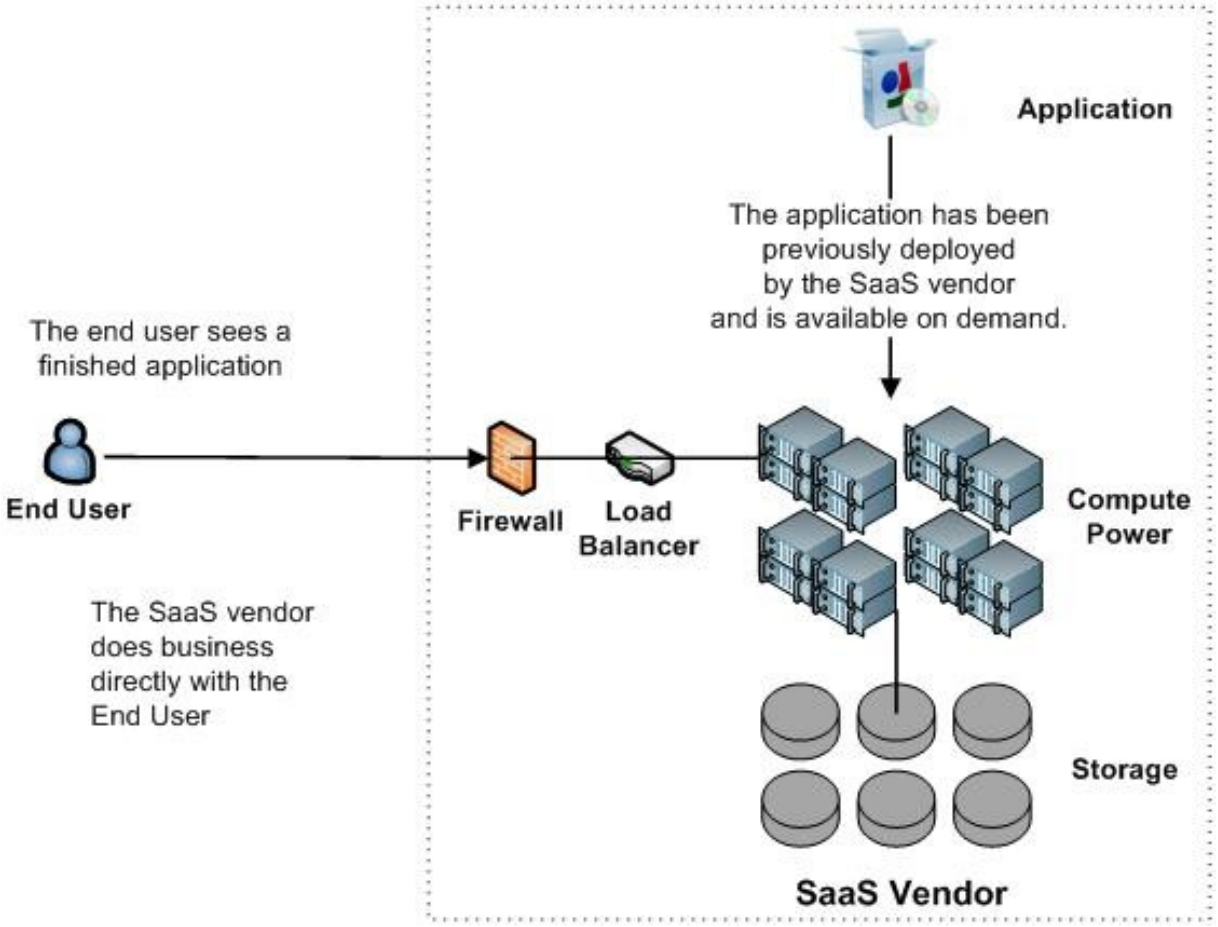
- Now


Public Cloud
The Cloud Provider
Public Cloud
The Cloud Provider
SME
Hybrid Cloud
Connectivity
(Network Access)
SME
Enterprise
Private Cloud
SME
Enterprise
Private Cloud



Architectural Layers of Cloud Computing

Software as a Service (SaaS)



salesforce.com
Success On Demand.

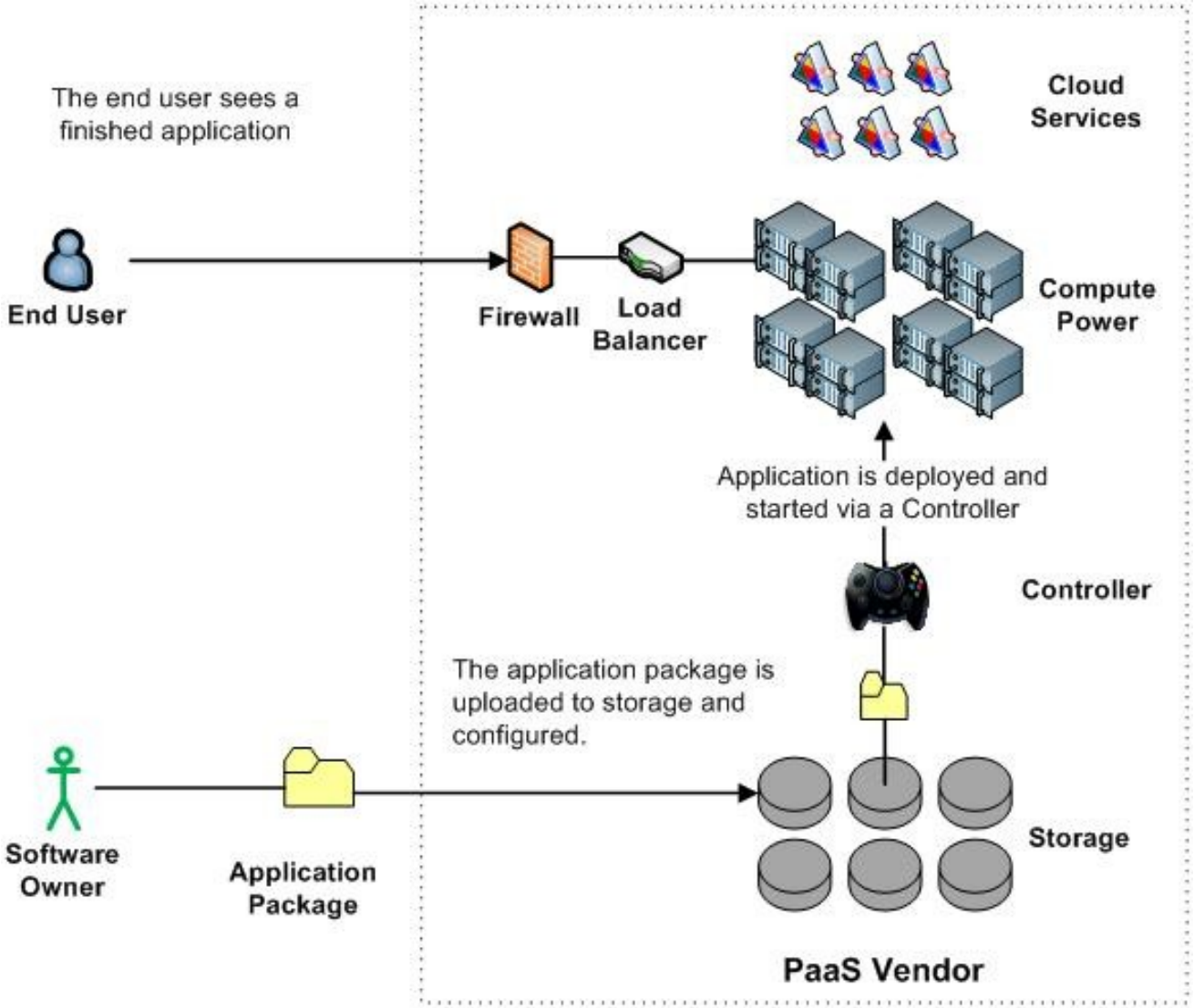
ZOHO
Work. Online

sliderocket

clarizen
Projects Made Real

Google Docs

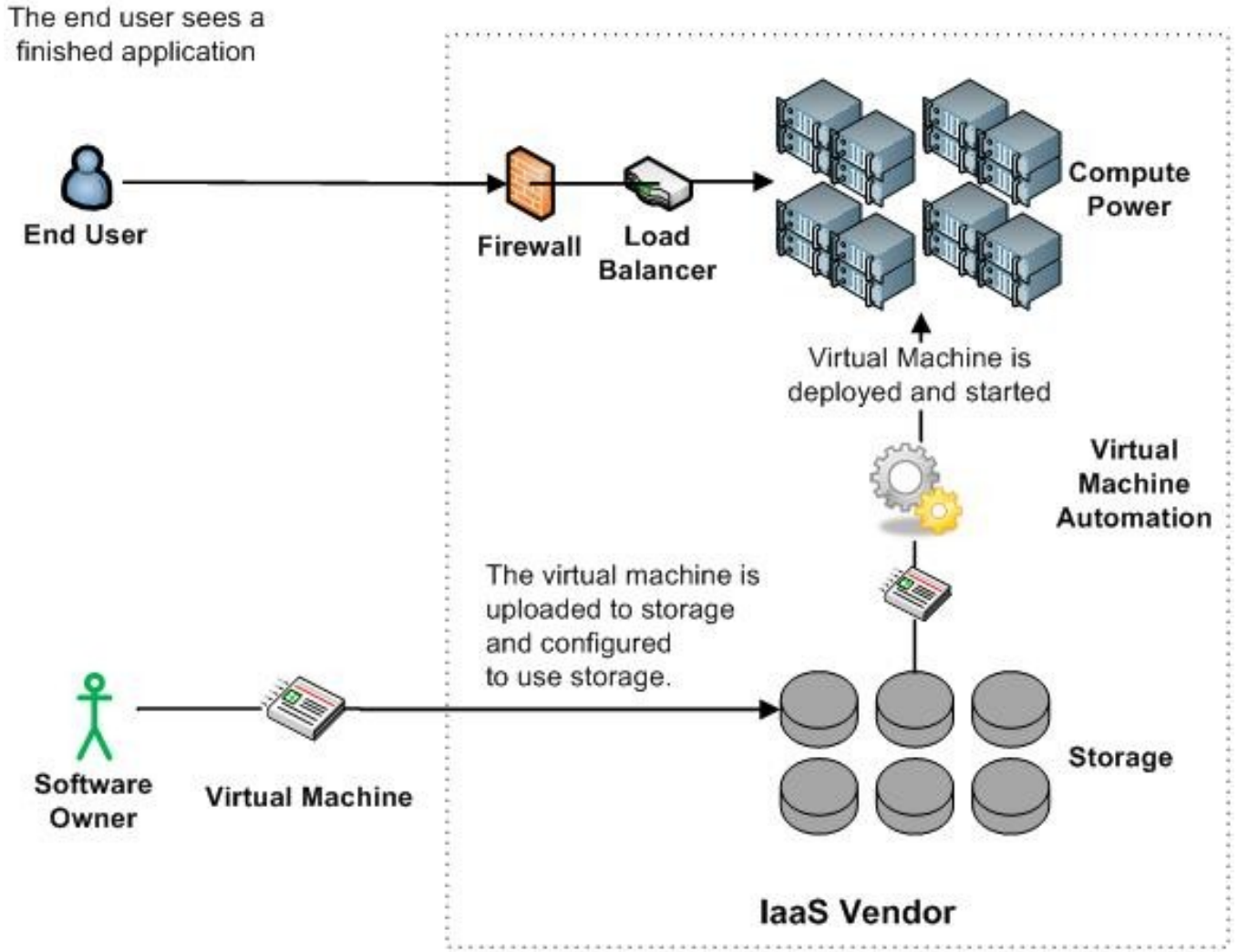
Platform as a Service (PaaS)



Google App Engine

Windows Azure

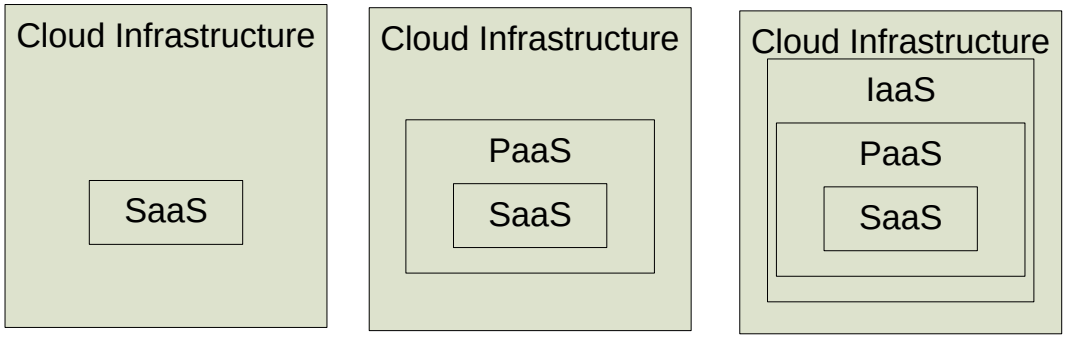
Infrastructure as a Service (IaaS)



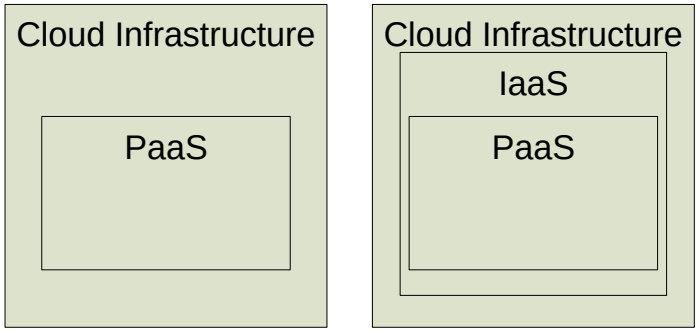
amazon web services™

rackspace

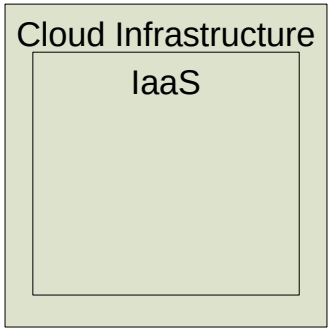
ServePath



Software as a Service (SaaS) Architectures



Platform as a Service (PaaS) Architectures



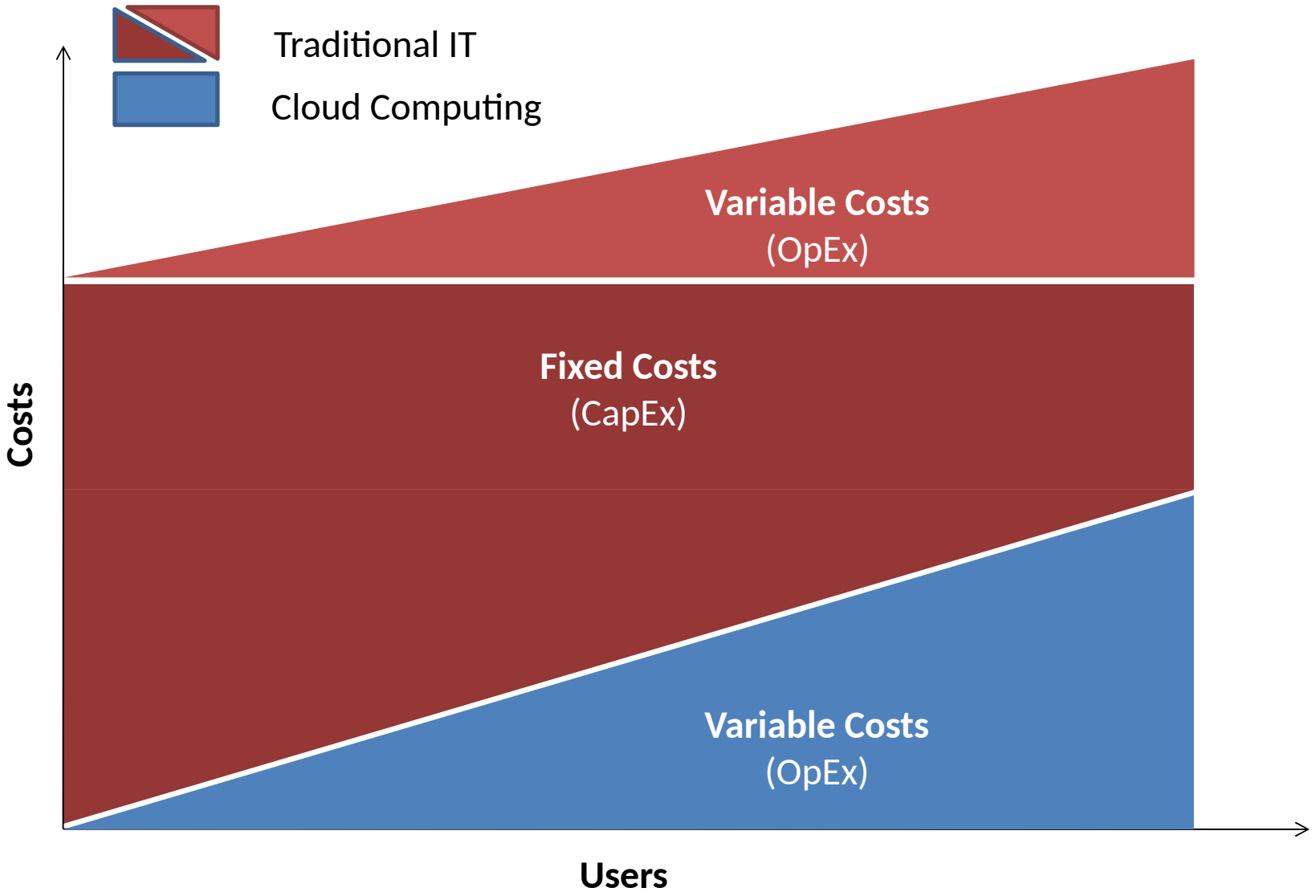
Infrastructure as a Service (IaaS) Architectures


Cloud Computing Characteristics

Comparisons

Grid Computing	Utility Computing	Autonomy Computing
• A form of distributed computing • A “super and virtual computer” is composed of a cluster of networked • Loosely coupled computers acting in concert to perform very large tasks	• Packaging of computing resources, such as computation and storage • A metered service similar to a traditional public utility, such as electricity	• Computer systems capable of self-management

Cloud Computing Economics



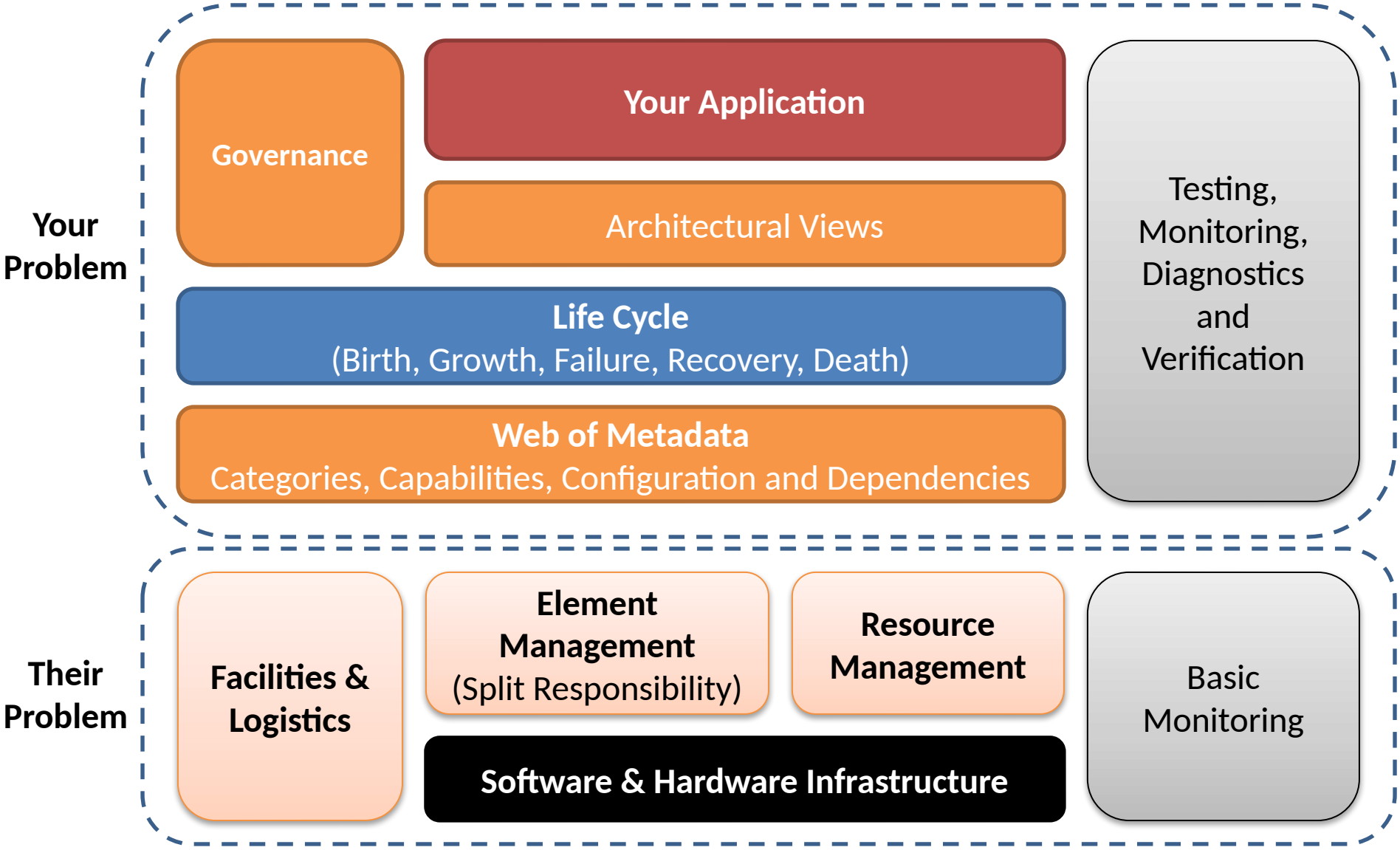
Pros and Cons



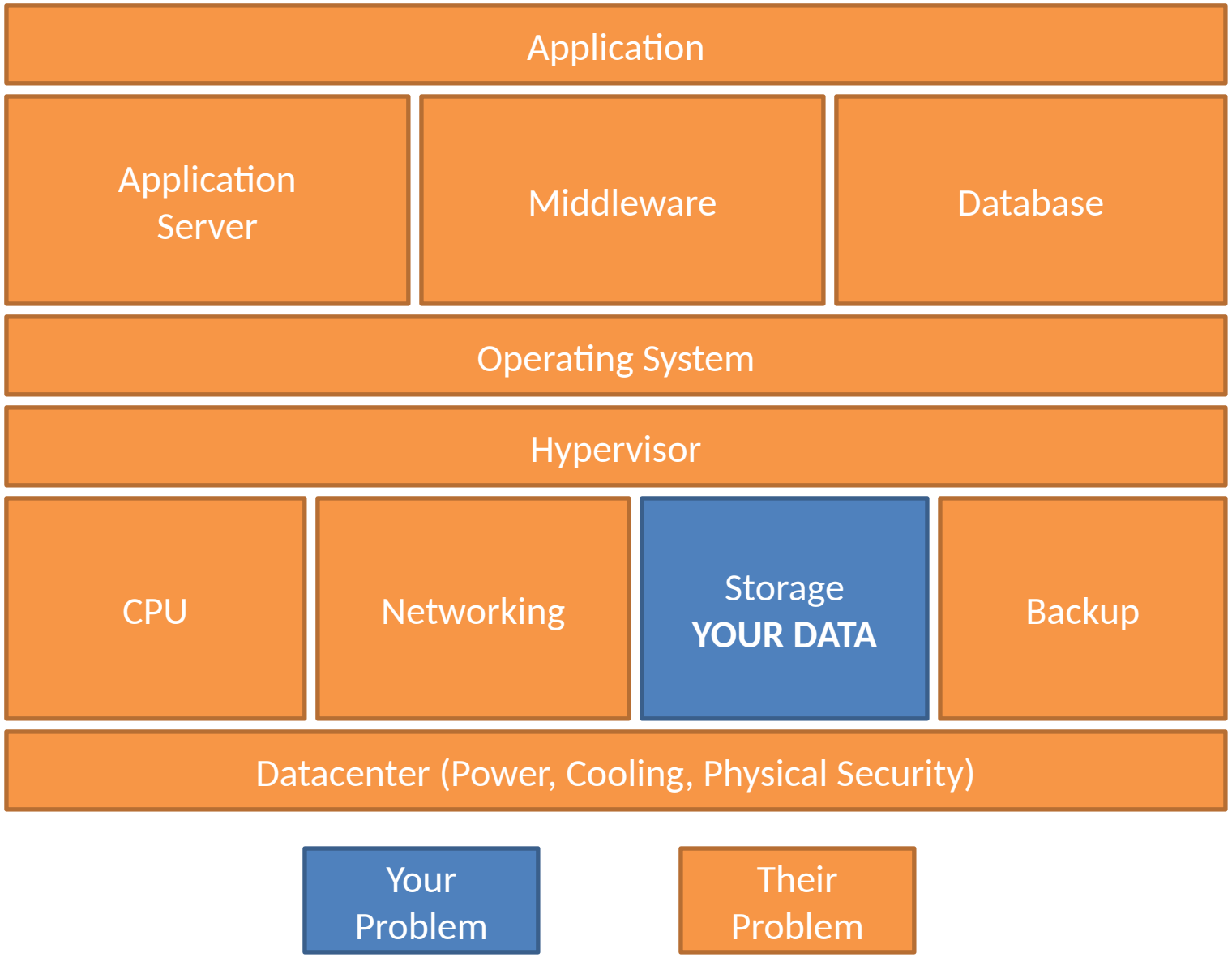


Cloud Computing Security

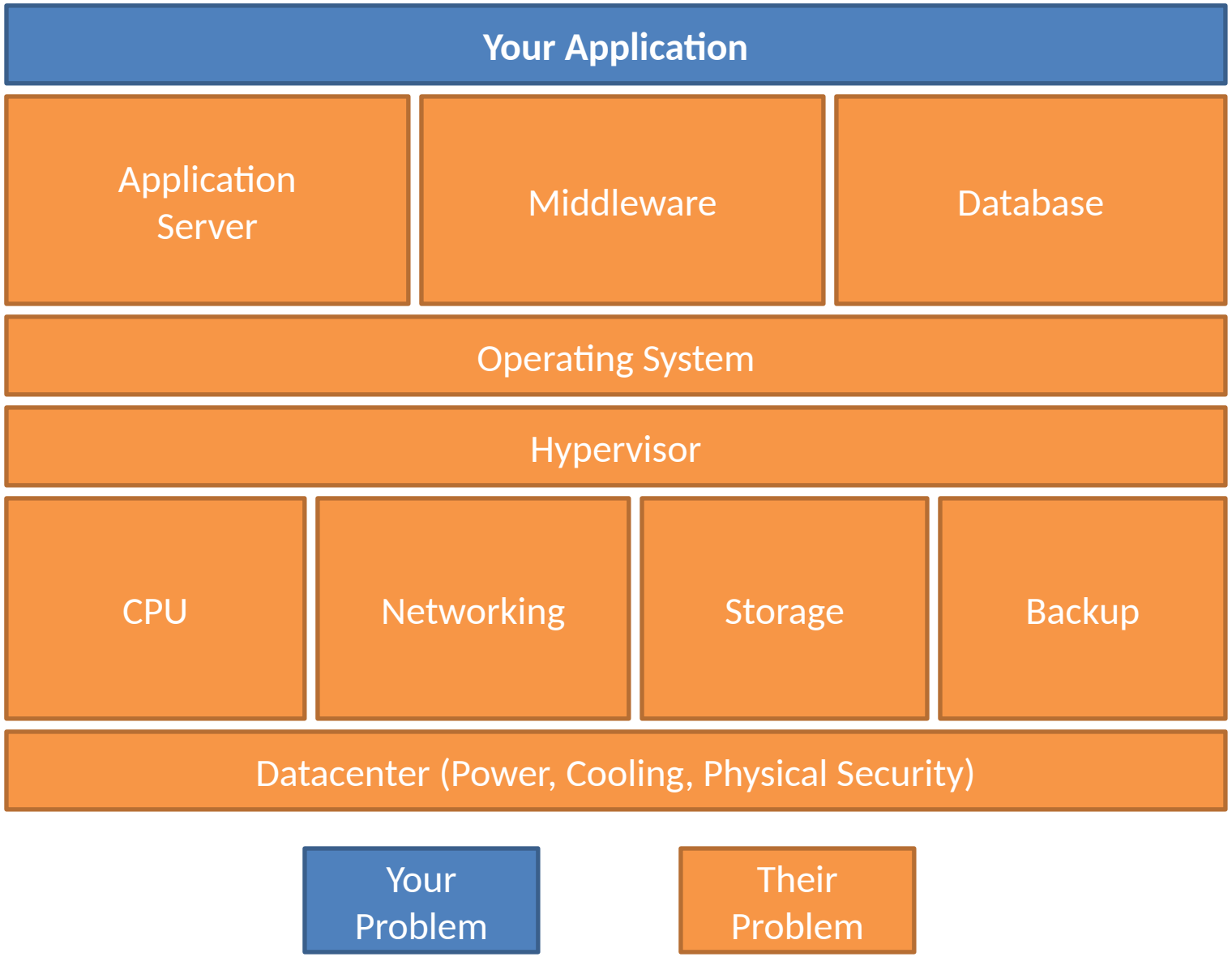
A Cloud Technology Reference Model



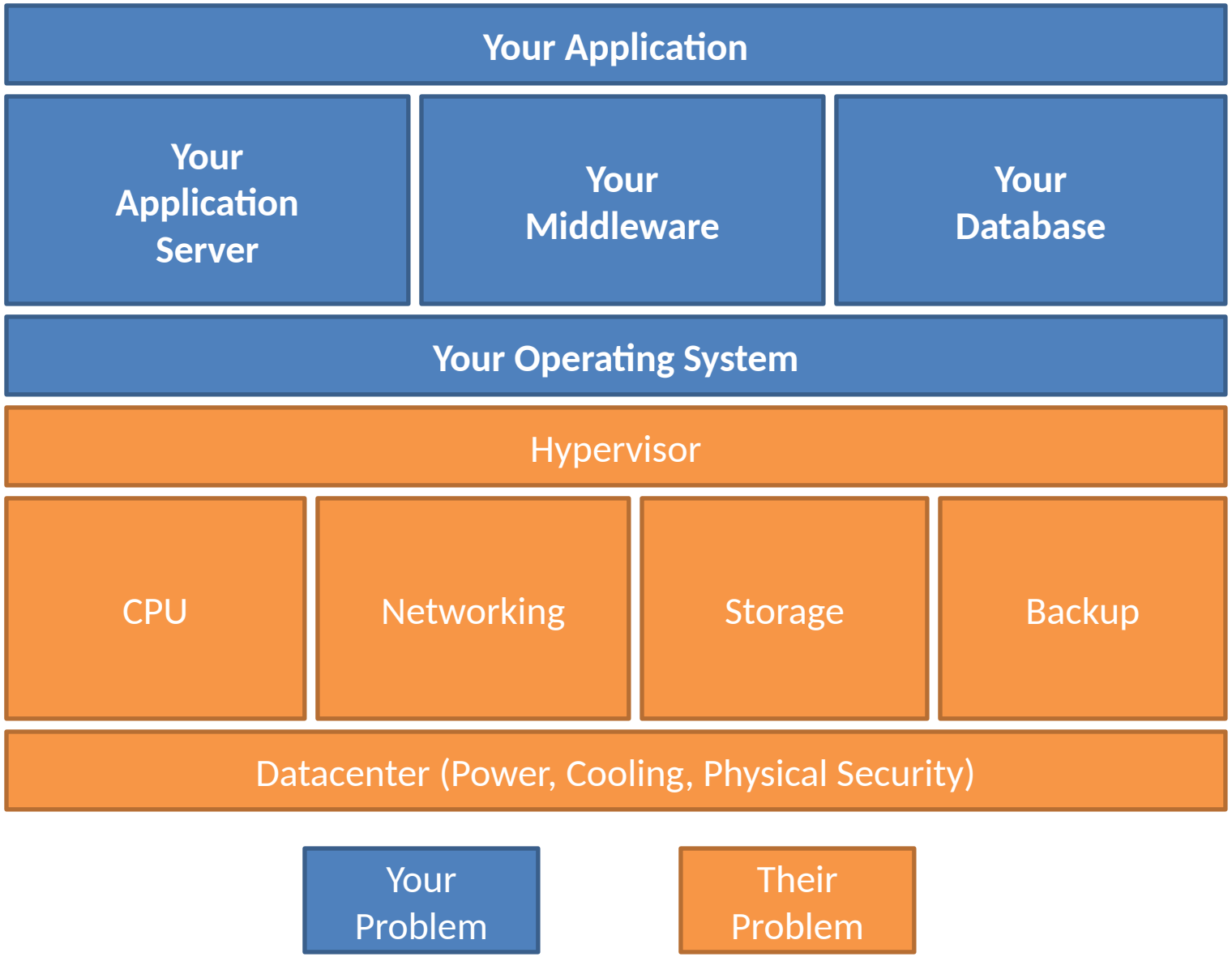
Software as a Service



Platform as a Service



Infrastructure as a Service





Security Issues

Overview

Governing in the Cloud

- Governance & Enterprise Risk Management
- Legal
- Electronic Discovery
- Compliance and Audit
- Information Life Cycle Management
- Portability & Interoperability

Operating in the Cloud

- Traditional Security
- Data Center Operations
- Incident Response
- Virtualization
- Identity & Access Management
- Storage
- Application Security
- Encryption & Key Management

Selected Issues

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Thank You